Exercise 6

On De Moivre's theorem

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Find the trigonometric form and the algebraic form of the solutions of the following equations and represent it on Argand diagram. Where $z \in \mathbb{C}$:**

(1) $z^3 = 1$ (2) $z^6 = -1$ (3) $z^3 = i$ (4) $z^3 = -i$

2 **Represent in Argand diagram :**

(1) The fifth roots of the number -32 (2) The fourth roots of the number $16\,i$

3 **Find the solution set in $\mathbb{C}$ of each of the following equations :**

(1) $z^4 = 16$ (2) $z^3 + 8 = 0$ (3) $z^5 - 32 = 0$ (4) $z^5 + 32\,i = 0$

4 **Find the trigonometric form of the following :**

(1) $\left(1 + \sqrt{3}\,i\right)^{\frac{1}{3}}$ (2) $\left(-\sqrt{3} + i\right)^{\frac{3}{5}}$ (3) $(1 - i)^{\frac{2}{5}}$ (4) $(-8)^{-\frac{1}{3}}$

5 **Find the square roots of each of the following numbers :**

(1) $7 - 24\,i$ (2) $\frac{3i - 1}{i + 3}$ (3) $\frac{8(1 - i)}{1 + i} + 15$

6 **Find the two square roots of the number z for each of the following :**

(1) *(2nd Session 2014)* $z = -8\,i$ in the exponential form.

(2) $z = 2 - 2\sqrt{3}\,i$ in the trigonometric form.

(3) *(2nd Session 2015)* $z = -3 + 4\,i$ without converting into trigonometric form.

7 Find in the trigonometric form and the exponential form the roots of the following in C : $z^4 = 8\left(1 - \sqrt{3}\,i\right)$ and then write its solution set.

8 **Find the trigonometric form and algebraic form of the two square roots for each of the following complex numbers :**

(1) $9(\cos 300° + i \sin 300°)$

(2) $4\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)$

(3) $\sin \frac{5\pi}{6} + i \cos \frac{5\pi}{6}$

(4) $4\,e^{\frac{5\pi}{3}i}$

9 If $z = 4 + 4\sqrt{3}\,i$ Find the exponential form of the number z , then find the cubic roots of the number z and represent it on Argand diagram.

10 ***(1st Session 2019)*** Put the number : $z = \frac{8}{1-\sqrt{3}\,i}$ on the trigonometric form then find its two square roots in the exponential form.

11 If $z = \sin\frac{\pi}{9} + i\cos\frac{\pi}{9}$, find $\bar{z}$ in the trigonometric form , then find the cubic roots of the number $(\bar{z})^9$

12 Find in simplest form : $z = \frac{1 + 4i + i^2}{2 - i - 2i^2 - i^3}$, then find the two square roots of the number z in the trigonometric form.

13 Find the modulus and the principle amplitude of the number : $\left(\frac{1+\sqrt{3}\,i}{1-i}\right)^{12}$, then find the two square roots of it in the exponential form.

14 Find in the different forms of the number : $z = \frac{-\sqrt{3}-i}{\sqrt{3}-i}$, then find the two square roots of the number z , then represent them in Argand diagram.

15 ***(1st Session 2012)*** If $z = \left(1 + \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^6$, put the number z in the trigonometric form , and then find the cubic roots of z in the exponenital form.

16 ***(1st Session 2009)*** If $z = \left(\frac{1+i}{1-i}\right)^5$, put the number z in the trigonometric form. Find the two square roots of the number z

17 $z = 6 - 8i$, then find $z^{\frac{1}{2}}$ in the algebraic form.

18 If $\mathcal{X} + y\,i = \frac{2}{1-\sqrt{3}\,i}$, find the value of each of the following : $\mathcal{X}$, y where $\mathcal{X}$, $y \in \mathbb{R}$, then find the square root of $\frac{2}{1-\sqrt{3}\,i}$ by algebraic method.

19 If $\frac{7-11\,i}{4+i} = a + b\,i$, find the value of $\left(\sqrt{-b} + a\,i\right)^{\frac{3}{2}}$

20 *(1st Session 2017)* If $z = 1 - \sqrt{3}\,i$ (where $i^2 = -1$)

Find $z^{\frac{3}{2}}$ in the trigonometric form.

21 If $z_1 = \sqrt{3}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$, $z_2 = 2(\sin 30 + i\cos 30)$, find $z_1 z_2$ in the exponential form , then determine the trigonometric form of the complex number z where $z = (z_1 z_2)^{\frac{1}{4}}$

22 *(2nd Session 2012)* If $z_1 = \left(\sin\frac{\pi}{9} + i\cos\frac{\pi}{9}\right)^5$, $z_2 = \left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)^4$, the complex number $z = \frac{z_1}{z_2}$, then find the two square roots of the complex number in the exponential form.

23 Put each of the following : $z_1 = 2\sqrt{3} + 2i$, $z_2 = 1 + \sqrt{3}\,i$ where $i^2 = -1$ in the exponential form , and then find the cubic roots of the number $z_1 z_2$

24 Put the number : $z_1 = -2 + 2\sqrt{3}\,i$ where $i^2 = -1$ in the exponential form and if : $z_1 z_2 = 16\,e^{\frac{11\pi i}{3}}$, then find two square roots of the number z_2 in trigonometric form.

25 If $z_1 = \frac{6+4i}{1+i}$, $z_2 = \frac{26}{5-i}$ where $i^2 = -1$, show that z_1 , z_2 are conjugate numbers then find the cubic roots of the number $4(z_1 - z_2)$ in the trigonometric form.

26 If $z_1 = \left(\frac{\sqrt{3}+i}{2}\right)^4$, $z_2 = \sin\frac{\pi}{3} + i\cos\frac{\pi}{3}$, $i^2 = -1$ and $z = \frac{z_1}{z_2}$, find the square roots of the number z in the trigonometric form.

27 *(1st Session 2000)* If $\mathcal{X} = \frac{4}{\sqrt{3}+i}$, $y = \frac{2}{\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}}$, prove that : $\mathcal{X}$, y are conjugate numbers , then find the cubic roots of the number z in the exponential form where $z = \mathcal{X}^2 - 2\mathcal{X}y + y^2$

28 If $z_1 = 1 - \sqrt{3}\,i$, $z_2 = \cos\theta + i\sin\theta$, $z_3 = \left(\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right)^2$ and $z = \frac{z_1 z_2}{z_3}$, find the modulus and principle amplitude of the number z , then find the two square roots of the number z in the trigonometric form at $\theta = \frac{\pi}{6}$

29 If $z_1 = 2\left(\sin\frac{\pi}{3} + i\cos\frac{\pi}{3}\right)$, $z_2 = \sqrt{2}\left(\sin\frac{\pi}{4} - i\cos\frac{\pi}{4}\right)$, $z_3 = 1 + \sqrt{3}\,i$

Find the number $z = \frac{z_1^3 \times z_2^4}{z_3^5}$ in the exponential form then find two square roots of the number z in the trigonometric form.

30 If $\sqrt{\frac{18+i}{2+3i}} = x + i y$ where $i^2 = -1$, x , $y \in \mathbb{R}$, find value of x , y

31 **Find the real values of x and y in each of the following :**

(1) $(x + i y)^2 = \frac{36 - 2i}{3 + 2i}$

(2) $(x + i y)^2 (1 + i) + 7 - i = 0$

(3) $(x + i y)^2 \left(1 + \sqrt{3}\, i\right) = -11 + 5\sqrt{3}\, i$

(4) $(x + i y)^2 - 10 (x + i y) + 41 = 0$

32 **Find the solution set for each of the following equation , where $x \in \mathbb{C}$:**

(1) $x^2 - (1 + i) x - 6 + 3 i = 0$

(2) $(1 - i) x^2 - (6 - 4 i) x + 9 - 7 i = 0$

(3) $x^2 i + (1 - 2 i) x + 3 i = 1$

33 **Choose the correct answer from the given ones :**

(1) The fifth roots of unity represented on Argand diagram vertices of .....

(a) an equilateral triangle. (b) a square.

(c) a regular pentagon. (d) a regular hexagon.

(2) If z_o , z_1 ... , z_5 represent the sixth roots of unity on Argand diagram then $m(\angle z_r O z_{r+1})$ = ... where $0 \le r \le 5$, $r \in \mathbb{Z}$

(a) 30° (b) 60° (c) 90° (d) 120°

(3) If $\sqrt{7 + 24 i} = x + iy$, then $(x + y)^2$ = ...

(a) 7 (b) 24 (c) 49 (d) 576

(4) $\sqrt{5 + 12 i}$ = ...

(a) $\pm (2 + 3 i)$ (b) $\pm (3 + 2 i)$ (c) $\pm (2 - 3 i)$ (d) $\pm (3 - 2 i)$

(5) If $z_1 = 2\left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}\right)$ and $z = -2i\,\overline{z_1}$, then one of the square roots for the number (z) is .....

(a) $\sqrt{2}\, e^{\frac{5\pi}{12} i}$ (b) $2e^{\frac{5\pi}{12} i}$ (c) $\sqrt{2} e^{\frac{-7\pi}{12} i}$ (d) $2e^{\frac{7\pi}{12} i}$

(6) *(1st Session 2024)* If $z = 8\, e^{\frac{\pi}{2} i}$, then one of the cubic roots of the complex number z is .....

(a) $\sqrt{3} + i$ (b) $\sqrt{3} - i$ (c) $\frac{1}{2} + \frac{\sqrt{3}}{2} i$ (d) $\frac{1}{2} - \frac{\sqrt{3}}{2} i$

(7) *(2nd Session 2024)* If the complex number $z = \sqrt{3} - 3 i$, then one of the cubic roots of the complex number z is .....

(a) $\sqrt[6]{12}\, e^{\frac{-5}{9}\pi i}$ (b) $\sqrt[6]{12}\, e^{\frac{\pi}{2} i}$ (c) $\sqrt[6]{12}\, e^{\frac{5}{9}\pi i}$ (d) $\sqrt[6]{12}\, e^{\frac{-\pi}{2} i}$

(8) If z_1 , z_2 , z_3 are the cubic roots of the number $z = 6 + 8\,i$, then $\frac{25\,z_1^3 \times 4\,z_2^6}{z_3^{12}} =$ .....

(a) $6 + 8\,i$ (b) $6 - 8\,i$ (c) $3 + 4\,i$ (d) $3 - 4\,i$

(9) The modulus of a complex number z is 16 and z_1 , z_2 are the square roots for the number z , then the length of the line segment whose end points z_1 and z_2 equals .....

(a) 32 (b) 16 (c) 8 (d) 4

(10) If z_1 , z_2 , z_3 are the roots of the equation $z^3 = 4 + 4\sqrt{3}\,i$, then the area of the polygon whose vertices represent z_1 , z_2 , z_3 on Argand diagram equals ..... square units.

(a) $3\sqrt{3}$ (b) $6\sqrt{3}$ (c) $9\sqrt{3}$ (d) $12\sqrt{3}$

(11) ***(1st Session 2020)*** The product of the roots of the equation : $x^4 - 1 = 0$ equals .....

(a) zero (b) 1 (c) – 1 (d) i

(12) If z_1 , z_2 , z_3 are the roots of the equation $z^3 = a$ where $a \neq$ zero and $\arg(z_1) = \theta$, then argument of z_2 , z_3 are ..... respectively.

(a) θ , θ (b) $\theta + \frac{\pi}{3}$, $\theta + \frac{2\pi}{3}$

(c) $\theta + \frac{2\pi}{3}$, $\theta + \frac{4\pi}{3}$ (d) $\frac{2\pi}{3}$, $\frac{4\pi}{3}$

(13) If a complex number (z) and its square roots lie on one circle , its centre at the origin , then which of the following is definitely known ?

(a) $|z|$ (b) arg (z)

(c) The real part of (z) (d) The imaginary part of (z)

(14) If z , z^2 , z^3 are complex numbers on Argand diagram and they are on one circle whose centre at the origin. Which of the following statements is true ?

(a) $\arg(z) = \arg(z^2) = \arg(z^3)$ (b) $|z| = 1$

(c) $|z| = |z^2| = |z^3| \neq 1$

(d) The triangle whose vertices are z , z^2 , z^3 is a right angled triangle.

(15) If z is a complex number , $z = r(\cos\pi + i\sin\pi)$, then the two roots of z are .....

(a) real and have same sign. (b) real and have different signs.

(c) imaginary and have same sign. (d) imaginary and have different signs.

(16) If z is a complex number , then the number of values of the number $z^{\frac{4}{3}}$ is.....

(a) one value. (b) two values. (c) 3 values. (d) 4 values.

(17) The sum of the two square roots of the complex number $(3 - 4\,i)$ equal .....

(a) zero (b) $\pm\left(\sqrt{3} - 2\,i\right)$ (c) $\pm(2 - i)$ (d) 6

(18) If $z_1 = 4\,e^{\frac{5\pi}{6}i}$, $z_2 = i$, then $\sqrt{z_1\,z_2}$ = ...

(a) $\pm\left(1 + \sqrt{3}\,i\right)$ (b) $\pm\left(1 - \sqrt{3}\,i\right)$ (c) $\pm\sqrt{3}\,i$ (d) $2 \pm \sqrt{3}\,i$

(19) The two square roots of the number i are .....

(a) $e^{\frac{\pi}{4}i}$, $e^{\frac{-3}{4}\pi i}$ (b) $e^{\frac{-\pi}{4}i}$, $e^{\frac{-3}{4}\pi i}$ (c) $e^{\frac{-\pi}{4}i}$, $e^{\frac{\pi}{4}i}$ (d) $e^{\frac{-3}{4}\pi i}$, $e^{\frac{3}{4}\pi i}$

(20) *(2nd Session 2022)* If z is a complex number and $z - 2 = i\,(z + 2)$, then the two square roots of the number z are ..... , where $i^2 = -1$

(a) $\pm(1 - i)$ (b) $\pm(1 + i)$ (c) $\pm(2 - i)$ (d) $\pm(2 + i)$

(21) *(1st Session 2022)* If the two square roots of the complex number $\frac{-7 + 26\,i}{5 - 2\,i}$ are $\pm(a + b\,i)$ where a , b are two real numbers , then $(a + 2\,b)$ could be equals .....

(a) 5 (b) 4 (c) zero (d) -4

(22) If $x + i\,y = i + \sqrt{i}$ where x and $y \in \mathbb{R}$, then $x + y$ = ...

(a) $1 + \sqrt{2}$ (b) $1 - \sqrt{2}$ (c) $1 \pm \frac{1}{\sqrt{2}}$ (d) $1 \pm \sqrt{2}$

(23) If $(1 - i)\,x + (1 + i)\,y + 2\,i = 0$, then the expression of $\sqrt{3\,x + 4\,y}\,i$ = ...

(a) $\pm(5 - i)$ (b) $\pm(2 - i)$ (c) $3 + i$, $1 - i$ (d) 6 , -5

(24) If z_1 , z_2 , z_3 , z_4 are roots of the equation $z^4 = a$ where $a \in \mathbb{R}^*$, then

First : polygon whose vertices are the points represent z_1 , z_2 , z_3 , z_4 on Argand plane represents .....

(a) rectangle. (b) square. (c) parallelogram. (d) trapezium.

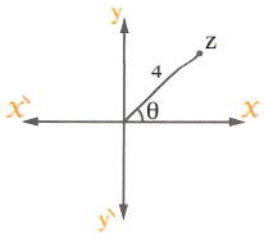
Second : If the area of this polygon = $4\sqrt{3}$ square unit , then a = ...

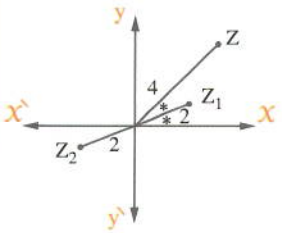
(a) 4 (b) 8 (c) 12 (d) 16

(25) If $z_1 = 2\,e^{\frac{2\pi}{3}i}$, $z_1\,z_2 = 8\,e^{\frac{11\pi}{3}i}$, then the product of the square roots of the number z_2 = ...

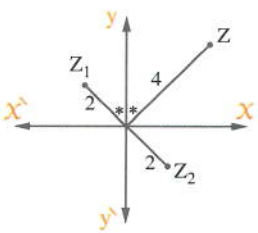
(a) $4\,e^{\pi i}$ (b) 4 (c) $4\,e^{\frac{\pi}{2}i}$ (d) 16

(26) If the opposite figure represents the complex number z. Which of the following figures does represent z_1 , z_2 , the two square roots of z ?

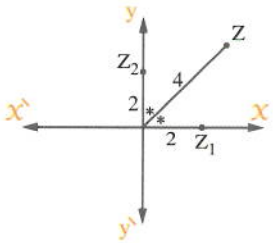




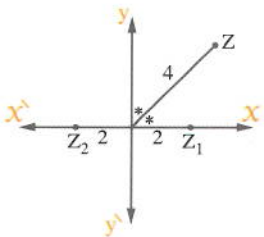
(a)



(b)



(c)



(d)

34 **Use De Moivre's theorem to express :**

(1) $\cos 3\theta$ in terms of powers of $\cos\theta$

(2) $\sin 3\theta$ in terms of powers of $\sin\theta$

35 **Use De Moivre's theorem to prove that :**

(1) $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$

$\sin 4\theta = 4(\sin\theta\cos^3\theta - \sin^3\theta\cos\theta)$

(2) $\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$

$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

36 **Prove that :** $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$

37 If $\sqrt[5]{x + y\,i} = a + b\,i$, prove that : $\frac{x}{a} - \frac{y}{b} = 4(b^4 - a^4)$